

Algorithms 2

Approximating Minimum Set Cover

Daniel Rosenberg & Michael Trushkin

Ariel University

1. LP Analysis for Greedy Set-Cover

Definition 1 Minimum cost set-cover problem

Given: a hypergraph $H := (V, E)$ and a cost function $c : E \rightarrow \mathbb{R}_{\geq 0}$.

Goal: find a hyperedge-cover $T \subseteq E$ with

$$\bigcup_{e \in T} e = V$$

minimizing

$$c(T) := \sum_{e \in T} c(e)$$

.

IP formulation (min cost SC):

Let x_e be an indicator variable for whether $e \in E$ is included in the cover. Then

$$\begin{aligned} & \min \sum_{e \in E} c(e)x_e \\ \text{s.t. } & \sum_{e \in C_v} x_e \geq 1, \quad \forall v \in V, \quad C_v := \{e \in E : v \in e\} \\ & x_e \in \{0, 1\}, \quad \forall e \in E \end{aligned}$$

LP relaxation:

$$\begin{aligned} & \min \sum_{e \in E} c(e)x_e \\ \text{s.t. } & \sum_{e \in C_v} x_e \geq 1, \quad \forall v \in V \\ & 0 \leq x_e \leq 1, \quad \forall e \in E \end{aligned}$$

Any solution with $x_e > 1$ can be reduced to $x_e = 1$ while preserving feasibility.

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LP relaxation:

$$\begin{aligned} & \min \sum_{e \in E} c(e)x_e \\ \text{s.t. } & \sum_{e \in C_v} x_e \geq 1, \quad \forall v \in V \\ & \cancel{0 \leq x_e \leq 1} \quad x_e \geq 0, \quad \forall e \in E \end{aligned}$$

Any solution with $x_e > 1$ can be reduced to $x_e = 1$ while preserving feasibility.

Dual of the LP relaxation:

$$\begin{aligned} & \max \sum_{v \in V} y_v \\ \text{s.t. } & \sum_{v \in e} y_v \leq c(e), \quad \forall e \in E \\ & y_v \geq 0, \quad \forall v \in V \end{aligned}$$

Integrality gap. Let $\text{OPT} := \text{OPT}(I)$ be the optimal IP value and $\text{OPT}_f := \text{OPT}_f(I)$ the optimal LP value. Then $\text{OPT}_f \leq \text{OPT}$, and

$$\min_I \frac{\text{OPT}(I)}{\text{OPT}_f(I)}$$

is the integrality gap.

Recall the greedy algorithm (weighted):

Algorithm 1: Greedy-SC

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1:  $A \leftarrow \emptyset$ 
2: while  $\bigcup A \neq V$  do
3:    $e \leftarrow \arg \min \{ \text{eff}_{\bigcup A}(e) : e \in E(H) \}$ 
4:   for each  $v \in e \setminus \bigcup A$  do
5:     Set  $\text{ac}(v) := \text{eff}_{\bigcup A}(e)$ 
6:   end
7:    $A \leftarrow A \cup \{e\}$ 
8: end
9: return  $A$ 

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Here $\text{eff}_{U(e)} := \frac{c(e)}{|e \setminus U|}$ and $\text{ac}(v)$ is the amortized cost of v .

Let $A = \{e_{i_1}, \dots, e_{i_m}\}$ be the greedy solution and $A_j = \{e_{i_1}, \dots, e_{i_j}\}$. Then

$$\text{cost}(A) = \sum_{j=1}^m c(e_{i_j}) = \sum_{j=1}^m |e_{i_j} \setminus \bigcup A_{j-1}| \cdot \frac{c(e)}{|e \setminus \bigcup A_{j-1}|} = \sum_{j=1}^m \sum_{v \in e_{i_j} \setminus \bigcup A_{j-1}} \text{eff}_{\bigcup A_{j-1}}(e_{i_j}) = \sum_{v \in V} \text{ac}(v)$$

Claim : Let $H_n := \sum_{i=1}^n \frac{1}{i}$. The vector $y_v := \frac{\text{ac}(v)}{H_n}$ for $v \in V$ is feasible for the dual.

Proof.

- Let $\{v_1, \dots, v_n\}$ be the order in which vertices are covered by the greedy algorithm.
- For any $e \in E$ with $|e| = k$, let $\{v'_1, \dots, v'_k\}$ be the order in which vertices of e are covered.
- When v'_i is covered, $|e \setminus \bigcup A| = k - i + 1$, so

$$\text{ac}(v_i) = \text{eff}_{\bigcup A}(e') \leq \text{eff}_{\bigcup A}(e) = \frac{c(e)}{|e \setminus \bigcup A|} = \frac{c(e)}{k - i + 1}.$$

- Hence

$$\sum_{v \in e} \text{ac}(v) \leq c(e) \sum_{i=1}^k \frac{1}{i} = H_k \cdot c(e)$$

- and

$$\sum_{v \in e} y_v = \sum_{v \in e} \frac{\text{ac}(v)}{H_n} \leq \left(\frac{H_k}{H_n} \right) c(e) \leq c(e).$$

The dual

$$\begin{aligned} & \max \sum_{v \in V} y_v \\ \text{s.t. } & \sum_{v \in e} y_v \leq c(e), \quad \forall e \in E \\ & y_v \geq 0, \quad \forall v \in V \end{aligned}$$

□

Claim : The greedy algorithm is an H_n -approximation for minimum cost set-cover.

Proof.

$$\text{cost}(A) = \sum_{v \in V} \text{ac}(v) = H_n \sum_{v \in V} y_v \leq H_n * \text{OPT}_f \leq H_n * \text{OPT}$$

□

Theorem 1 Weak Duality Theorem

$$\max\{c^T x : Ax \leq b\} \leq \min\{y^T b : A^T y = c, y \geq 0\}$$

whenever both LPs are feasible and bounded.

2. Set-Cover: Deterministic Rounding of the Primal LP

IP formulation

$$\min \sum_{e \in E} c(e)x_e$$

$$\text{s.t. } \sum_{e \in C_v} x_e \geq 1, \quad \forall v \in V$$

$$x_e \in \{0, 1\}, \quad \forall e \in E$$

LP relaxation

$$\min \sum_{e \in E} c(e)x_e$$

$$\text{s.t. } \sum_{e \in C_v} x_e \geq 1, \quad \forall v \in V$$

$$x_e \geq 0, \quad \forall e \in E$$

- For $v \in V(H)$, set $f_v := |C_v|$ (number of hyperedges containing v).
- Define $f := \max_{v \in V(H)} f_v$ (max frequency). Trivially $f \geq 1$.

Algorithm (deterministic rounding):

- Solve the LP relaxation optimally and let $(x_e^*)_{e \in E(H)}$ be the solution.
- Return $A := \left\{ e \in E(H) : x_e^* \geq \frac{1}{f} \right\}$.

Theorem 2 The algorithm above forms an f -approximation for minimum cost set-cover.

Proof.

- The set A is a valid cover: for any $v \in V(H)$,
- $\sum_{e \in C_v} x_e^* \geq 1$ implies $\exists e \in C_v$ with $x_e^* \geq \frac{1}{|C_v|} \geq \frac{1}{f}$.

- Define

$$z_e := \begin{cases} 1 & e \in A \\ x_e^* & e \notin A \end{cases}$$

- Then $z_e \leq f x_e^*$ for all e . (if $e \in A$, then $x_e^* \geq \frac{1}{f}$)
- Hence

$$c(A) \leq \sum_{e \in E(H)} z_e c(e) \leq f \sum_{e \in E(H)} x_e^* c(e) = f \cdot \text{OPT}_f \leq f \cdot \text{OPT}.$$

□

3. Set-Cover: Randomised Rounding of the Primal LP

IP formulation

$$\min \sum_{e \in E} c(e)x_e$$

$$\text{s.t. } \sum_{e \in C_v} x_e \geq 1, \quad \forall v \in V$$

$$x_e \in \{0, 1\}, \quad \forall e \in E$$

LP relaxation

$$\min \sum_{e \in E} c(e)x_e$$

$$\text{s.t. } \sum_{e \in C_v} x_e \geq 1, \quad \forall v \in V$$

$$x_e \geq 0, \quad \forall e \in E$$

Randomised algorithm:

- Solve the LP relaxation optimally and let $(x_e^*)_{e \in E(H)}$ be the solution.
- Include each $e \in E(H)$ independently with probability x_e^* to solution A .
- Return A .

Lemma 3 Let A be the collection returned by the algorithm and fix $v \in V(H)$. Then $\mathbb{P}[v \in \bigcup A] \geq 1 - \frac{1}{e}$.

Proof.

- Let $C_v := \{e \in E(H) : v \in e\}$.
- Then

$$\begin{aligned} \mathbb{P}[v \notin \bigcup A] &= \mathbb{P}[\text{no } e \in C_v \text{ is chosen}] \\ &= \prod_{e \in C_v} (1 - x_e^*) \\ &\leq \prod_{e \in C_v} \exp(-x_e^*) = \exp\left(-\sum_{e \in C_v} x_e^*\right) \leq e^{-1} \end{aligned}$$

.

□

Conclusion: $\mathbb{P}[A \text{ is not a valid cover}] \leq \sum_{v \in V(H)} \mathbb{P}[v \notin \bigcup A] \leq \frac{n}{e}$.

Looks bad!

- For $k \in \mathbb{N}$, run the algorithm k independent times to obtain $A_1, A_2, \dots, A_k$.
- Set $U_k := \bigcup_{i=1}^k A_i$ (union of all solutions).
- For $v \in V(H)$, $\mathbb{P}[v \notin U_k] \leq (e^{-1})^k = e^{-k}$.

Lemma 4 There exists a constant $D > 0$ such that if $k \geq D \ln n$, then $\mathbb{P}[U_k \text{ is not a set-cover}] \leq \frac{1}{4}$.

Proof.

- For a fixed $v \in V(H)$, $\mathbb{P}[v \notin U_k] \leq e^{-k} \leq e^{-D \ln n} \leq \frac{1}{4n}$ for some D .
- A union bound over all vertices gives $\mathbb{P}[U_k \text{ not a set-cover}] \leq \frac{1}{4}$.

□

Lemma 5 Let OPT be the optimal cost and $k \in \mathbb{N}$. Then $\mathbb{P}[c(U_k) \geq 4 \cdot k \cdot \text{OPT}] \leq \frac{1}{4}$.

Proof.

- For $e \in E(H)$, set

$$X_e := \begin{cases} 0 & e \notin A \\ c(e) & e \in A \end{cases}$$

- Then $\mathbb{E}[X_e] = c(e) \cdot \mathbb{P}[e \in A] = c(e)x_e^*$.
- Since $c(A) = \sum_{e \in E(H)} X_e$,

$$\mathbb{E}[c(A)] = \sum_{e \in E(H)} \mathbb{E}[X_e] = \sum_{e \in E(H)} x_e^* c(e) = \text{OPT}_f$$

.

- Therefore $\mathbb{E}[c(U_k)] \leq k \cdot \text{OPT}_f$, and by Markov's inequality,

$$\mathbb{P}[c(U_k) \geq 4 \cdot k \cdot \text{OPT}] \leq \frac{k \cdot \text{OPT}_f}{4 \cdot k \cdot \text{OPT}} \leq \frac{1}{4}$$

□

Conclusion. Repeating the algorithm $k := \Omega(\ln n)$ times and taking U_k gives with probability at least $\frac{1}{2}$ a valid cover with cost of at most $O(\ln n * \text{OPT})$.

- For $k = \Omega(\ln n)$,

$$\mathbb{P}[U_k \text{ is a cover} \wedge c(U_k) \leq 4 \cdot k \cdot \text{OPT}] \geq 1 - \mathbb{P}[U_k \text{ not a cover}] - \mathbb{P}[c(U_k) \geq 4 \cdot k \cdot \text{OPT}] \geq \frac{1}{2}$$

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